

NYCU-EE IC LAB – Fall 2025

Lab06 Exercise

Design: WinRate & Poker

Data Preparation

Extract files from the TA directory:
`% tar xvf ~iclabTA01/Lab06.tar`

Texas Hold'em Poker Rules

■ Objective

The goal of Texas Hold'em is to win chips (or points) by:

- Having the **best 5-card poker hand** at showdown, OR
- Forcing all other players to fold their cards (No need to consider this situation in this lab).

■ Setup

- Each player is dealt **two private cards** (called *hole cards*).
- Five **community cards** are placed face-up in the center of the table. All players share these cards.
- Each player makes the best possible **5-card hand** by combining their two hole cards with the five community cards (any mix: 0, 1, or 2 hole cards can be used).

■ The Four Betting Rounds

1. **Pre-flop** – After receiving hole cards, players bet.
2. **Flop** – The dealer reveals the first three community cards; another betting round follows.
3. **Turn** – A fourth community card is revealed; another betting round.
4. **River** – The fifth and final community card is revealed; final betting round.

■ Showdown

- If more than one player remains after the last betting round, all players show their cards.
- The **best 5-card hand** wins the pot.

■ Poker Hand Rankings (Highest → Lowest)

1. **Royal Flush** – A-K-Q-J-10, all of the same suit.
2. **Straight Flush** – Five consecutive cards of the same suit. (A-K-Q-J-10 is the best, A-2-3-4-5 is the weakest. For example, K-Q-J-10-9 is better than J-10-9-8-7.)
3. **Four of a Kind** – Four cards of the same rank.
4. **Full House** – Three of a kind + a pair.

Design Description

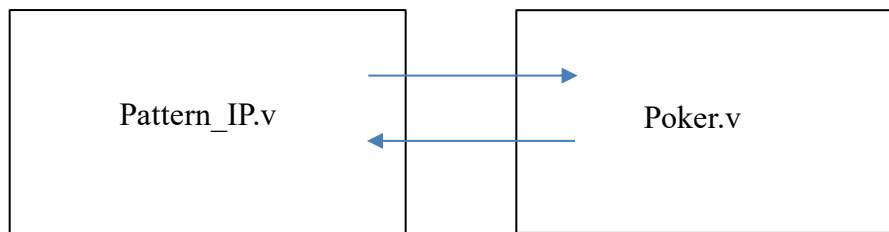
Design	Definition
Soft IP(Poker.v)	Given each player's hole cards and the 5 community cards, the Soft IP computes the best 5-card poker hand for each player, and outputs winner.
Top Design(WinRate.v)	The Top Design receives inputs (9 players' cards and 3 community cards), and use the Soft IP to count and outputs 9 players' win rate.

According to the above introduction, you need to design your own soft IP to determine the winner, and use it to calculate win rate in the top design.

The summary of both design in this Exercise are as above:

■ *Soft IP*

- Input each player's 2 hole cards and the 5 community cards.



Inputs & Outputs (Poker.v)

Input Signal	Bit Width	Description
IN_HOLE_CARD_NUM	8*IP_WIDTH	MSB to LSB : {card1_num [3:0], card2_num[3:0] }, from player [N] to player[0].
IN_HOLE_CARD_SUIT	4*IP_WIDTH	MSB to LSB : {card1_suit[1:0], card2_suit [1:0] }, from player [N] to player [0].
IN_PUB_CARD_NUM	20	MSB to LSB : {card1_num [3:0], card2_num[3:0], card3_num [3:0], card4_num [3:0], card5_num [3:0] }, 5 public card number.
IN_PUB_CARD_SUIT	10	MSB to LSB : {card1_suit[1:0], card2_suit [1:0], card3_suit[1:0], card4_suit[1:0], card5_suit[1:0] }, 5 public card suit.

Card_num	2	3	4	5	6	7	8	9	10	11	12	13	14
Poker_num	2	3	4	5	6	7	8	9	10	J	Q	K	A

Suit_num	0	1	2	3
Poker_suit	Clubs	Diamonds	Hearts	Spades

Output Signal	Bit Width	Description
OUT_WINNER	IP_WIDTH	If the player N win the game, please pull the Nth bit to high. For example, if there are total 6 player and player 2 and player 5 win, you will output"100100".

Specifications (Soft IP)

Top module

1. Top module name: **Poker**(design file name: **Poker.v**)
2. Finish calculating within **one** clock cycle.
3. You need to use **generate** to design this soft IP.
4. **The look-up table method is forbidden.** (TA will check your design)

Supplement

5. Don't use any wire/reg/submodule/parameter name called *error*, *congratulation*, *latch* or *fail* otherwise you will fail the lab. Note: * means any char in front of or behind the word. e.g: error_note is forbidden.
6. Don't write chinese comments or other language comments in the file you turned in.
7. Verilog commands //synopsys dc_script_begin, //synopsys dc_script_end
//synopsys translate_off, //synopsys translate_on are only allowed during the usage of including and setting designware IPs, other design compiler optimizations are forbidden.
Using the above commands are allowed, however any error messages during synthesize and simulation, regardless of the result will lead to failure in this lab.
Any form of display or printing information in verilog design is forbidden. You may use this methodology during debugging, but the file you turn in should not contain any coding that is not synthesizable.

■ Top Design

- Input **9 player's 2 hole cards** and the **3 community cards** on Flop.
- You are required to calculate the probability that each player will win the hand after the turn and river cards are dealt, given only the three community cards on the flop.
- Express in percentage and truncate to the integer part, and the every players' win rate percentage before truncated should be total 100%.



■ Inputs & Outputs (WinRate.v)

Input Signal	Bit Width	Description
clk	1	Clock.
rst_n	1	Asynchronous active-low reset.
in_valid	1	High when input signals are valid.
in_hole_num	72	MSB to LSB : {card1_num [3:0], card2_num[3:0] }, from player[8] to player[0].
in_hole_suit	36	MSB to LSB : {card1_suit[1:0], card2_suit [1:0] }, from player[8] to player[0].
in_pub_num	12	MSB to LSB : {card1_num [3:0], card2_num[3:0], card3_num [3:0] }, 3 public card number.
in_pub_suit	6	MSB to LSB : {card1_suit[1:0], card2_suit [1:0], card3_suit[1:0] }, 3 public card suit.

Output Signal	Bit Width	Description
out_valid	1	High when input signals are valid.
out_win_rate	63	MSB to LSB : {player[8]win_rate[6:0], player[7]win_rate[6:0], ...}

	<pre>player[1]win_rate[6:0], player[0]win_rate[6:0]}</pre> Express in percentage and truncate to the integer part	
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Specifications (WinRate.v)

Top module

1. Top module name: **WinRate** (design file name: **WinRate.v**).
2. You can adjust your clock period by yourself, but the maximum period is **20ns**. The precision of clock period is 0.1, for example, 4.5 is allowed, but 4.55 is not allowed.
3. The execution latency is limited to **1,000 cycles**. The latency is the clock cycles between the falling edge of the last cycle of **in_valid** and the rising edge of the **out_valid**.
4. The total cell area should not be larger than **1000,000 um²**.
5. **The look-up table method is forbidden**, and you need to **use your own-designed soft IP (Poker) in the top module**. (TA will check your design)

Reset

6. It is **asynchronous** reset and **active-low** architecture. If you use synchronous reset (considering reset after clock starting) in your design, you may fail to reset signals.
7. The reset signal(**rst_n**) would be given only once at the beginning of simulation. **All output signals should be reset after the reset signal is asserted**.

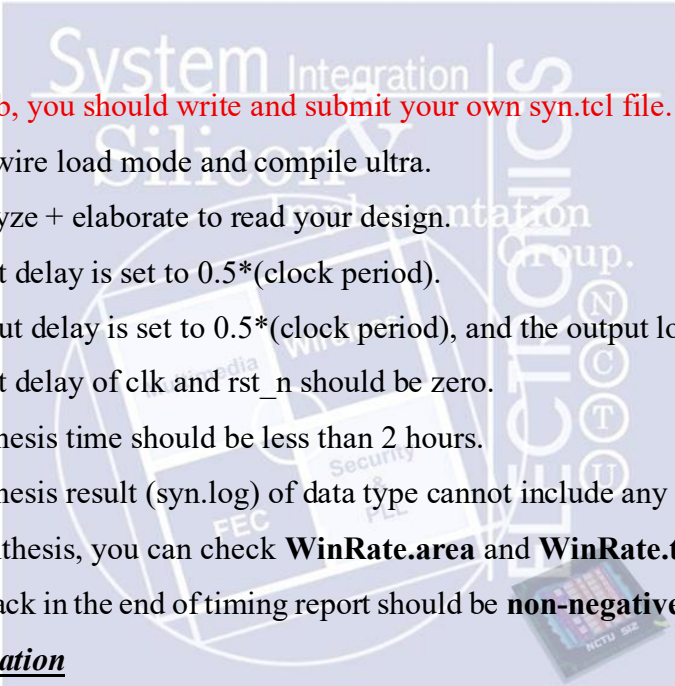
in_valid

8. **in_valid** will come after reset.
9. All input signals are synchronized at **negative edge** of the clock.
10. When **in_valid** is low, input is tied to unknown state.

out_valid

11. **out_valid** should not be raised when **in_valid** is high.
12. The **out_valid** is limited to be high for only **1 cycles** when the output value is valid.
13. All output signals should be synchronized at clock positive edge.
14. The TA's pattern will capture your output for checking at **clock negative edge**.
15. The **out_win_rate** should be correct when **out_valid** is high.
16. The **out_win_rate** should be reset after your **out_valid** is pulled down.
17. The next input pattern will come in **2~6 negative edge of clock** after your **out_valid** falls.

Synthesis

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18. In this lab, you should write and submit your own syn.tcl file.
 19. Use **top** wire load mode and compile ultra.
 20. Use analyze + elaborate to read your design.
 21. The input delay is set to 0.5*(clock period).
 22. The output delay is set to 0.5*(clock period), and the output loading is set to 0.05.
 23. The input delay of clk and rst_n should be zero.
 24. The synthesis time should be less than 2 hours.
 25. The synthesis result (syn.log) of data type cannot include any latches and error.
 26. After synthesis, you can check **WinRate.area** and **WinRate.timing**. The area report is valid only when the slack in the end of timing report should be **non-negative** and the result should be **MET**.

Gate level simulation

27. The gate level simulation cannot include any timing violations without the notimingcheck command.

Supplement

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Grading Policy

1. Grading policy:

- Function Validity: 50% (**WinRate.v** RTL and **Gate-level** simulation correctness)
 - ◆ **IP_WIDTH = 9**
- Performance: 30% (Area * Cycle Time * Execution Latency Cycle)
- Soft IP function correctness: 20% (RTL simulation Only, no second demo)

◆ **IP_WIDTH = 2 ~ 9**

- The performance is determined by **area** and **latency** of **WinRate.v**. The less cost your design has, the higher grade you get.
- There are many effective algorithms for both soft IP and top design. You can select your preferred one for better performance.
- The grade of 2nd demo would be **30%** off.

Note

1. Submit your design at **Lab06/09_SUBMIT**
2. Submit your design through **Lab06/09_SUBMIT/01_submit**
 - a. 1st_demo deadline: **2025/10/20(Mon.) 11:59:59**
 - b. 2nd_demo deadline: **2025/10/22(Wed.) 11:59:59**
3. If your file **violates the naming rule**, you will **lose 5 points**.
4. Don't use any wire/reg/submodule/parameter name called ***error***, ***congratulations***, ***latch*** or ***fail*** otherwise you will fail the lab. Note: * means any char in front of or behind the word.
e.g: error_note is forbidden.

In RTL simulation, the name of template folder and reference commands is:

1. 01_RTL (RTL simulation):
./01_run_vcs_rtl
2. 02_SYN/ (Synthesis):
./01_run_dc_shell
(Check **latch** by searching the keyword "**Latch**" in 02_SYN/syn.log)
(Check the design's timing in /Report/ WinRate.timing)
(Check the design's area in /Report/ WinRate.area)
3. 03_GATE/ (Gate-level simulation):
./01_run_vcs_gate
4. **09_SUBMIT/ (submit your files):**
./00_tar
./01_submit
./02_check

You can key in **./09_clean_up** to clear all log files and dump files in each folder

Example Waveform

Input and output signal:

